Standard Deviation

- Multiple trials (e.g., by different network initializations)
  - Multiple performance scores $\{a_i\}_{i=1:N}$
- Calculate the mean and standard deviation
  - $\hat{\mu} = \frac{\sum_{i=1}^{N} a_i}{N}$
  - $\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^{N} (a_i - \hat{\mu})^2}{N}}$
- Report standard deviation in tables
- Draw error bars in the plots with the standard deviation

Significance Test

- Issue: Given two models, a set of performance scores on each, how confidently we can say one is better than the other?
- Answer: We make statistical significance test *of the mean difference*
- Null Hypothesis: samples of performance scores were drawn from the same distribution
- Reject the null hypothesis: a p-value

Significance Test

Common situation

- Two models have been tested N times
  - e.g. N random trials for proposed approach vs baseline
  - e.g. using N-fold cross-validation
- The performance scores are based on same data for both models
- Use the mean of N scores to represent the summary statistics of each model
- Assume that the sample mean (and possibly scores) of each model are Gaussian-distributed
- Use paired Student's t-test

Paired Student t-test

- simple steps

1. Calculate the paired differences $d_i = a_i - b_i$
2. Calculate the mean difference $\hat{d} = \frac{\sum d_i}{n}$
3. Calculate the std of the difference σ_d
4. Calculate t-statistics: $\mathrm{T} = \frac{\hat{d}\sqrt{n}}{\sigma_d}$
5. Use look-up table to get the p-value for how well it follows the t-distribution

Standard Deviation

Standard Error of the Mean

- Multiple (N) trials (e.g., by different network initializations)
  - Multiple performance scores $\{a_i\}_{i=1:N}$
- Treat the mean ($\hat{\mu} = \frac{\sum_{i=1}^{N} a_i}{N}$) of these performance scores as a random variable.
- Calculate the standard deviation ($\sqrt{\frac{\sum_{i=1}^{N}(a_i-\hat{\mu})^2}{N}}$)
- Calculate the variance of the mean
  - $Standard\ Error\ (\mathrm{SE}) = \sqrt{\mathrm{var}[\frac{\sum_{i=1}^{N} a_i}{N}]} = \sqrt{\frac{\mathrm{var}[a_i]}{N}} \approx \frac{\sqrt{\frac{\sum_{i=1}^{N}(a_i-\hat{\mu})^2}{N}}}{\sqrt{N}}$
- Report 95% confidence intervals in tables as numbers
  - 95% interval = [$\hat{\mu}$ - SE*1.96, $\hat{\mu}$ + SE*1.96]
- Draw error bars in the plots corresponding to the 95% confidence interval

In any case be crystal clear about what it is that you are reporting!

Probably good enough, but usually statistically “incorrect” due to many reasons…

Significance Test

the choice of significance test algorithm

- What performance measure is used?
  - classification accuracy
  - mean absolute error
  - …
- What data was used for obtaining performance samples?
  - Same data for both models (paired test)
  - Or different data (unpaired test)

Significance Test: issues of common approach

the choice of significance test algorithm

- What is our assumption about distribution of each model's sampled performances?
  - Gaussian with same variance (e.g. student t-test)
  - Gaussian with different variance (e.g. Welch's t-test)
  - Non-Gaussian (e.g. Kolmogorov-Smirnov test)
- How do we represent the population-level performance of each model?
  - Mean or median usually
  - Does the test takes this into account?